# Introduction to Artificial Intelligence

For accounting students | 2 hours 30 minutes

**Learn the Basics**

**Use AI Wisely**

**Build with Agents**

https://www.linkedin.com/in/amnuay

GitHub QR code (to be added)

GitHub repository

Presenter: Mr. Amnuay Mekchompu, B.Eng, B.Acc, MBA

# What is AI?

## Simple definition

- The science and engineering of making smart machines
- Experts do not agree on one exact definition
- Routine tasks often stop being called AI

## Is machine learning different from AI?

- AI is the broad goal: machines doing tasks that need human-like intelligence
- Machine learning is a part of AI. It learns patterns from data
- Rule-based AI follows rules that people write by hand

## How computers understand different inputs

### Sentences

The text is cut into tokens, and each token becomes a list of numbers (see slide 5).

### Images

An image is a grid of pixels, each a number for brightness. Models learn shapes and edges from many labeled pictures.

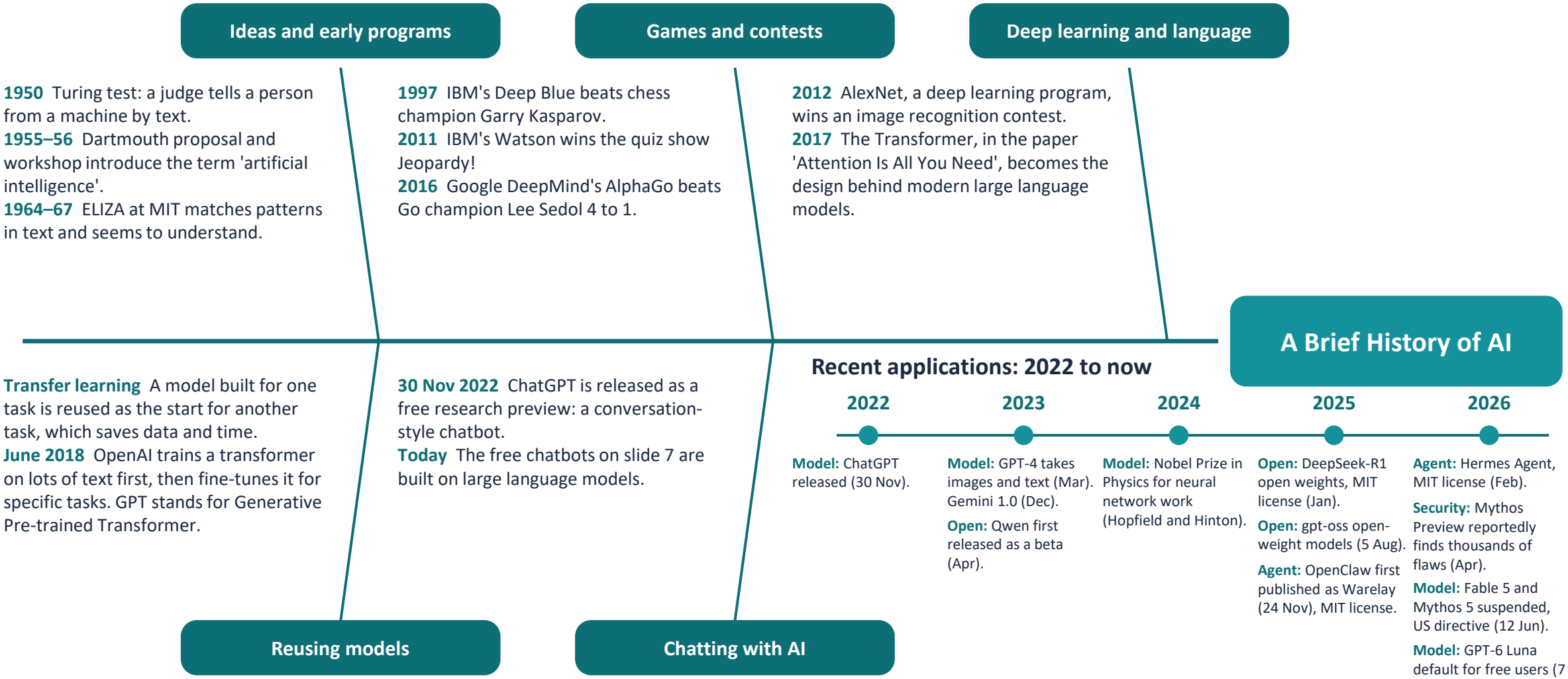
### Voice (speech)

Sound is turned into features, and a model picks the most likely words. Language models help choose the word order.

**Games: AI plays chess and Go at a high level (see slide 3).**

Sources (full links in speaker notes): John McCarthy, Stanford; Glasgow University; IBM (machine learning, computer vision, speech recognition); Quanta Magazine; Jay Alammar.

# A Brief History of AI




Sources (full links in speaker notes): Stanford Encyclopedia of Philosophy (Turing), Wikipedia (Dartmouth, ELIZA, Qwen, OpenClaw, Hermes Agent), arXiv (Transformer, DeepSeek-R1), OpenAI, Google (Gemini), Nobel Prize (2024), GitHub (DeepSeek-R1), Anthropic (Glasswing, Fable and Mythos), TechTarget.

# Key terms you will hear

### AI
Machines doing tasks that need human-like intelligence.

### Machine Learning
Computers learn patterns from data instead of fixed rules.

### Model
A trained program that finds patterns and makes outputs.

### LLM (Large Language Model)
AI trained on huge amounts of text to understand and write.

### GPT
Generative Pre-trained Transformer: the design behind ChatGPT.

### Chatbot
An app you chat with, built on top of an LLM.

### Prompt
The message or question you type to the AI.

### Instructions
Standing rules the AI follows in every message of a chat.

### Context
What the AI can see in one chat: your messages and files.

### Memory
Facts kept across chats, such as your preferences.

### Effort
How long and hard the AI thinks before it answers.

### Hallucination
A confident answer that is wrong or made up.

### AI Agent
AI that plans steps and uses tools to reach a goal.

### Tool
An action the AI can take, such as searching the web.

### Skill
A folder of instructions that teaches the AI a task.

Sources (full links in speaker notes): Google Cloud, Symphony AI, E2E Networks, Lenovo, Coddy, Gradually AI, Benton.org, Msty docs, Coram AI, IBM, Stanford University.

# How an LLM reads text: tokens and vectors

## 1. Tokens: text is cut into pieces

I | love | account | ing

Example split (illustration). Real splits depend on the model.

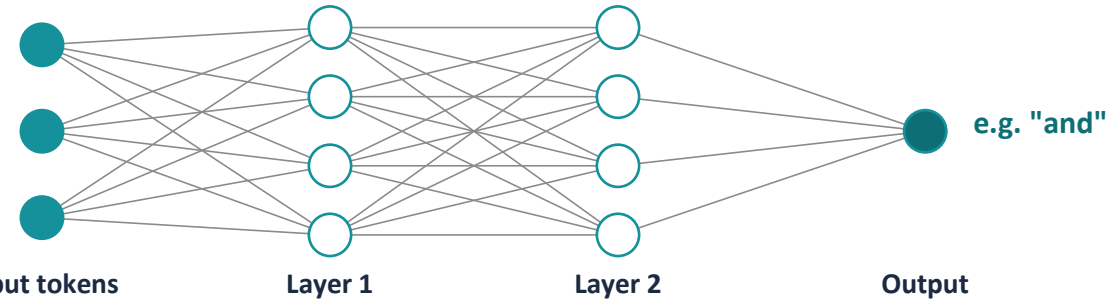
## 2. Vectors: each token becomes a list of numbers

account → [ 0.12, −0.87, 0.45, … ]

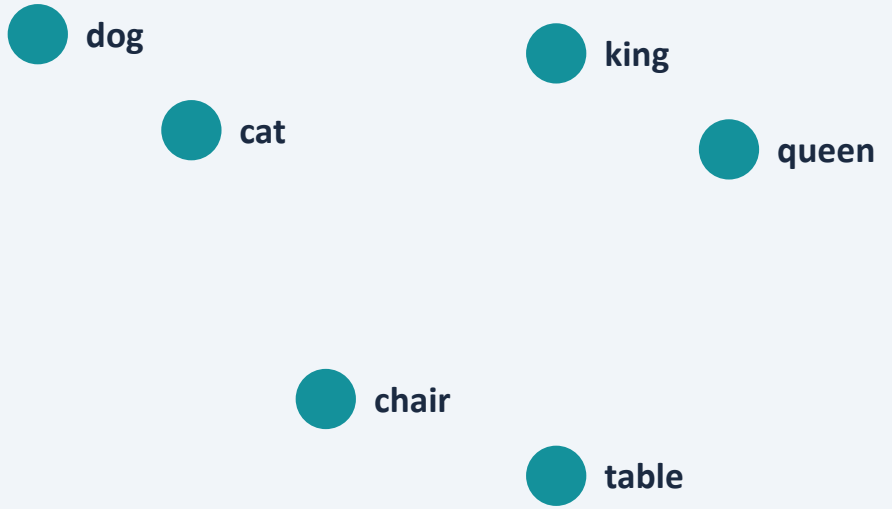
Example numbers (illustration). Llama 3 uses 4,096 numbers per word; GPT-3 uses 12,288.

## 3. Prediction: the model guesses the next token

Illustration: circles hold numbers; lines are connections with learned numbers (weights).




## 3. Meaning as distance




Words used in similar ways sit close together. This map uses 2 dimensions. Real models use thousands.

Sources: Quanta Magazine, how embeddings encode meaning, https://www.quantamagazine.org/how-embeddings-encode-what-words-mean-sort-of-20240918 | Gupta Deepak glossary (token), https://guptadeepak.com/guides/glossary/context-window/ | Jay Alammar, The Illustrated Transformer, https://jalammar.github.io/illustrated-transformer/

# What LLMs are good at, and where they fail

## Good at

- Summarizing, translating and drafting emails, articles and reports
- Answering questions about text you paste in
- Works 24 hours a day and does not get tired
- Handles digital text much faster than people for many tasks

## Limits

- Can give misleading or wrong answers with confidence
- Can repeat bias when training data is poor
- Forgets parts of a long chat that no longer fit

## Get better answers and fewer mistakes

### Instructions

Set the role and rules once, so every answer follows them.

### Effort (thinking level)

Higher effort can give more careful answers on hard tasks. It is slower and can cost more.

### Prompt engineering

Writing clear prompts: be specific, allow 'I don't know', ask for quotes from your document.

Sources (full links in speaker notes): Anthropic (two pages), TechTarget, Coram AI, Lenovo, Gradually AI.

# AI tools with free plans

## ChatGPT

Made by OpenAI

**Free plan: yes**

Model: GPT-6 Luna is the default for free users (OpenAI, Oct 2026)

Best for: General tasks; widest free features

## Gemini

Made by Google

**Free plan: yes**

Model: Gemini family; flagship is Gemini 4 Argon. Free model access changes

Best for: Google users and long documents

## Microsoft Copilot

Made by Microsoft

**Free plan: yes**

Model: OpenAI GPT-5.6 plus Microsoft's own MAI models

Best for: Web search and Microsoft users

## Perplexity

Made by Perplexity AI

**Free plan: yes**

Model: Own Sonar models (Sonar, Sonar Pro, Sonar Reasoning)

Best for: Research with cited sources

## Claude

Made by Anthropic

**Free plan: yes**

Model: Fable 5.1, Opus 5.5, Sonnet 5.5, Haiku 5.5

Best for: Writing and working with documents

Frontier model: a general-purpose AI that is among the most capable available. Example: Google calls Gemini 4 Argon its flagship model.

Open weight: the trained model files can be downloaded and run by anyone (for example OpenAI's gpt-oss). Open source adds the right to study and change it.

Free plans and limits change often. Check each website before class. Free plans may use your chats to improve AI.

Sources (full links in speaker notes): OpenAI release notes, Anthropic, Google, Microsoft (via TechCrunch), Perplexity (via Artificial Analysis), DataCamp, OSI, OpenAI gpt-oss.

# Using AI well: four controls

## Instructions

- Set the role and rules once
- Example: 'You are a tutor. Explain in simple English.'

## Context

- Give the facts the AI needs in this chat
- Bigger is not always better: models can miss details in the middle of long inputs

## Memory

- Kept across chats, separate from context
- You can review, delete, or turn it off

## Effort

- More effort gives more careful answers
- It is slower and can cost more; level names differ by app

**Try this: ask the AI how its memory works in the app you use. Then say what to keep and what to ignore.**

Sources: Coddy, https://coddy.tech/docs/prompts/system-prompt | Gradually AI, https://www.gradually.ai/en/ai-glossary/context-window/ | Tom's Guide, https://tomsguide.com/ai/chatgpt-now-has-a-memory-heres-what-it-means | Benton.org, https://www.benton.org/headlines/make-ai-remember-you | Alma, https://alma.now/docs/features/reasoning | Msty, https://docs.msty.studio/features/thinking-effort

# Using AI safely

## Check facts

- Check numbers by hand, especially money and tax figures
- Open the sources an AI cites before you trust them

## Human in the loop

- A person reviews and approves important steps
- Errors can build up across steps

## Protect your data

- Free plans may use your chats to improve AI; check the settings
- Do not paste private client data unless your organization allows it

## Where your data can stay: on-premise options (running AI on hardware you control)

### Own computer

Apps such as Ollama run open-weight models on macOS, Windows or Linux. Ollama says its open models can be used with complete privacy, locally or in the cloud.

### Desktop AI computer

NVIDIA DGX Spark has 128 GB of memory and can run AI models of up to 200 billion parameters. Price varies by seller.

### Organization server

A company or faculty can run its own AI servers, with its IT team managing security. Teaching suggestion, not sourced.

Sources (full links in speaker notes): IBM, aiweekly, Dupple, Ollama, NVIDIA. Client-data and organization-server items are teaching suggestions.

# Break

10 minutes.  Back at: _____

# What is an AI Agent?

## Definition

- An AI system that uses a language model and tools to complete a goal
- It plans steps, acts, checks results, and repeats until the goal is met or it cannot continue

## Normal chatbot

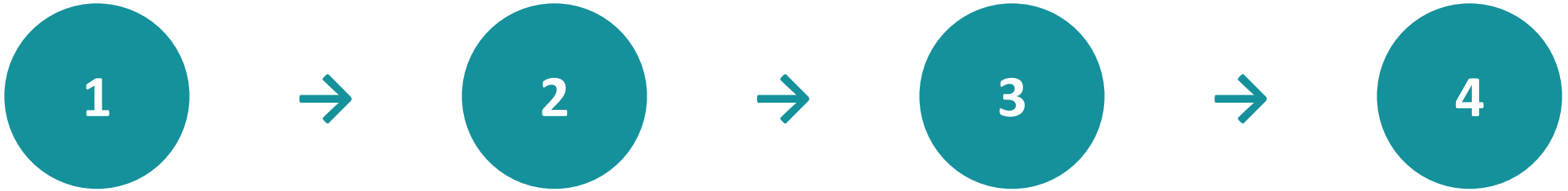
- Waits for your message
- Answers once, then stops
- Has no tools

## AI agent

- Takes a goal
- Plans steps and uses tools
- Checks results and reports back

Sources: IBM, https://www.ibm.com/think/topics/ai-agents | aiweekly, https://aiweekly.co/learning-ai/glossary/what-is-an-ai-agent

# How an agent works: the loop




### Goal

You describe the task

### Plan

It breaks the task into steps

### Act

It uses a tool, such as web search

### Check

It reviews the result and repeats if needed

## Watch out

- Errors can build up across steps
- Good agents keep human checkpoints before important actions

Sources: Mastra, https://mastra.ai/articles/ai-agents | aiweekly, https://aiweekly.co/learning-ai/glossary/what-is-an-ai-agent | IBM, https://www.ibm.com/think/topics/ai-agents

# What are Skills?

### What a skill folder looks like

```
my-skill/
    SKILL.md      main instructions
    scripts/      optional code
    templates/    optional files
```

### How the agent uses a skill

- 1. It sees only each skill's name and short description
- 2. It loads the full skill when the task needs it
- 3. It follows the steps and uses any helper files
- Use skills only from trusted sources: a malicious skill can cause harm

Sources: Anthropic engineering, https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills | inference.sh, https://inference.sh/blog/skills/agent-skills-overview

# Tools vs Skills vs Agents

## Tool

- One action the AI can take
- Example: search the web

## Skill

- Instructions for doing a task well
- Example: a checklist for a bank reconciliation

## Agent

- Plans steps and uses tools and skills
- Example: prepare a reconciliation from uploaded files, then wait for your approval

**MCP (Model Context Protocol): a standard way to connect AI to outside tools and software.**

Sources: IBM, https://www.ibm.com/think/topics/ai-agents | Anthropic, https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills. Examples are teaching ideas, not from a source.

# Sample questions and trick tests

## Sample questions (teaching suggestions)

Explain depreciation to a first-year accounting student in five short bullets.

อธิบายค่าเสื่อมราคาให้นักศึกษาปีหนึ่งที่เรียนบัญชีฟัง ใน 5 ข้อสั้นๆ

Act as a tutor. Explain debit and credit in simple English, with one example.

ทำหน้าที่เป็นอาจารย์พิเศษ อธิบายเดบิตและเครดิตด้วยภาษาง่ายๆ พร้อมตัวอย่าง 1 ข้อ

What is the difference between a balance sheet and an income statement?

งบแสดงฐานะการเงิน (Balance Sheet) กับงบกำไรขาดทุน (Income Statement) ต่างกันอย่างไร

Summarize this text in three bullets: [paste a short, non-private paragraph]

สรุปข้อความนี้เป็น 3 ข้อสั้นๆ: [วางย่อหน้าสั้นๆ ที่ไม่ใช่ข้อมูลส่วนตัว]

Make a five-question quiz on double-entry bookkeeping. Put the answers at the end.

ทำแบบทดสอบ 5 ข้อเรื่องการบัญชีคู่ (Double-entry) และใส่เฉลยไว้ท้ายสุด

Name one common student mistake in VAT calculations. Give a source I can check.

บอกข้อผิดพลาดที่นักศึกษามักทำเรื่อง VAT 1 ข้อ พร้อมแหล่งอ้างอิงให้ตรวจสอบ

## Trick questions: AI often gets these wrong. Test them yourself.

Trick: I want to wash my car at a car wash 200 meters away. Should I walk there or drive?

ฉันอยากเอารถไปล้างที่คาร์แคร์ห่าง 200 เมตร ควรเดินไปหรือขับไป

Trick: How many r's are in the word strawberry?

คำว่า strawberry มีตัวอักษร r กี่ตัว

Trick: Which is bigger, 9.11 or 9.9?

ระหว่าง 9.11 กับ 9.9 อันไหนมากกว่ากัน

How to test on OpenRouter (openrouter.ai/chat): 1. Choose a model from the list. 2. Paste one trick question. 3. Record the model name, the answer, and whether it passed.

**Free models to test (IDs from OpenRouter): openai/gpt-oss-20b:free, openai/gpt-oss-120b:free, nvidia/nemotron-3-ultra-550b-a55b:free, minimax/minimax-m3.1-flash:free, thinkingmachines/inkling:free**

Sources (full links in speaker notes): OpenRouter free model pages. Sample questions and trick questions are teaching suggestions.

# Live demo 1: chat, one step at a time (15 minutes)

| Step | Type this | Change to look for |
|---|---|---|
| **Start: check the context** | What context do you have in this chat right now? List the messages and files you can use.<br>ตอนนี้ คุณมี context อะไรบ้างในแชทนี้ ช่วยแสดงรายการข้อความและไฟล์ที่คุณใช้ได้ | A short list of this chat only. No pasted text yet. Keep it as your 'before' result. |
| **Step 1: plain question** | What is depreciation? | A long general answer in any layout. Note its length and how it is arranged. |
| **Step 2: add Instructions** | You are an accounting tutor. Answer in five short bullets. What is depreciation? | The answer becomes five short bullets. Compare it with Step 1. |
| **Step 3: add Context** | Use only the text below to explain depreciation: [paste a short, non-private paragraph] | The answer uses words from your paste. Run the context check again: the list now includes the pasted text. |
| **Step 4: compare Effort** | Ask the same question at a low and a high Effort setting, if your app has one. | The high setting may give more detail and take longer. Note the time and the detail. |
| **Step 5: check one fact** | Give a source link for each point, and say which page it came from. | Links appear. Open one. Does it say what the answer says? If not, mark it wrong. |

Sources (full links in speaker notes): Gradually AI (context window), Anthropic (extended thinking: more time). Prompts are teaching suggestions.

# Live demo 2: an AI agent checks a rule

## 1. The task: give the agent this goal (same goal for every pair)

**Goal: "Find the current VAT (value-added tax) rate in Thailand. Give the official Revenue Department page link, and the date shown on that page."**

## 2. Watch the four stages, and check what to look for at each one

### 1. Plan

Turns the goal into a research plan

**Watch for: Does the plan say to use an official government site?**

### 2. Search

Searches and opens web pages

**Watch for: Is the link on rd.go.th, the Revenue Department site?**

### 3. Reason

Reviews what it found and picks next steps

**Watch for: Did it check the date on the page?**

### 4. Report

Writes a report with findings

**Watch for: Does the report give the page link and its date?**

## 3. Check table: students open each link and fill this in

| Claim from the report | Link opens? (Yes / No) | Official site? (Yes / No) | Matches the page? (Yes / No) |
|---|---|---|---|
| | | | |
| | | | |
| | | | |

Official page to check: Revenue Department press releases, https://rd.go.th/295.html (full links in speaker notes).

# AI and the accounting profession

## 5 days

Faster annual report filing, on average, for firms that use more AI in accounting

## Under 3%

Share of accounting job postings in each area that are AI-related roles

## 3 themes

Routine tasks, data analysis, and value-adding professional roles

### Where AI is used in accounting work

**Bank reconciliation**
Matching bank lines to the ledger. Can shorten cycles from days to hours.

**Invoice processing**
Reading invoice data, matching it to purchase orders, and routing approvals.

**Document analysis**
Pulling key facts from contracts and flagging items for a person to review.

**Bookkeeping**
Less time on routine entries, so staff can focus on advice.

**Reporting**
Faster monthly statements. Some AI-using accountants finish within two weeks of month-end.

**Audit and fraud checks**
Reviewing transaction histories for unusual patterns and possible irregularities.

**AI is unlikely to fully replace accountants. Human judgment is still needed for complex decisions.**

Sources (full links in speaker notes): UNC Kenan-Flagler, RePEc concept paper, Economic Research journal, EFAA guide on AI in accounting.

# Ethics: using AI responsibly

UNESCO (United Nations Educational, Scientific and Cultural Organization) adopted the first global standard on AI ethics in November 2021. All 193 member states took part in approving it.

### Human rights and dignity

- The standard aims to protect human rights and human dignity

### Transparency and accountability

- UNESCO lists these as essential principles for AI

### Avoid bias

- Without guardrails, AI can repeat real-world bias and discrimination

Sources: UNESCO via Teaching and Learning Ireland, https://hub.teachingandlearning.ie/resource/recommendation-on-the-ethics-of-artificial-intelligence/ | UNESCO key facts, https://gcedclearinghouse.org/en/node/129379 | UNESCO brochure, https://www.unesco.nl/sites/default/files/inline-files/Unesco%20AI%20Brochure.pdf

# Summary and questions

## Key takeaways

- AI is the big field; machine learning and LLMs sit inside it
- A model is the trained result; instructions, context, memory and effort shape its answers
- Always check important answers and sources
- Agents plan and use tools to reach goals, with human checkpoints
- Skills teach agents how to do specific tasks

# Questions?

Thank you